

(need to change)Project 2 Linear Mixed Model

** **

March 15, 2026

1 Abstract

2 Introduction

3 Exploratory Data Analysis

An exploratory data analysis was conducted, aiming to understand the distributional properties of the heat flow measurements and identify preliminary trends regarding inter-individual and intra-individual variability, as well as the relationships between the different sensors.

3.1 Dataset Overview

The dataset consists of measurements from 24 participants (12 female, 12 male) monitored over two separate days. The only environmental difference between Day 1 and Day 2 is the temperature: Day 1 is 24 °C while Day 2 is 18 °C. While the original experiment captured data every second, this analysis utilizes the mean heat flow per activity in each session monitored by 6 sensors to address the research goals.

98 out of 2736 observations have missing heat flow measurements, and were directly dropped.

3.1.1 Nikita note to add: Specifically, the activity walking only has data for sessions 1 & 3, and the activity standing does not have data for session 2 on day 2.

All variables are categorical: `day` has 2 levels (“1” and “2”); `session` has 4 levels (“1”, “2”, “3” and “4”); `activity` has 3 levels (“sitting”, “standing” and “walking”); `sex` has 2 levels (“F” and “M”); `HF_type`(sensor ID) has 6 levels (“HF_2”, “HF_3”, “HF_5”, “HF_7”, “HF_8”, “HF_16”).

3.1.2 Nikita note to add: Note sensors correspond with XYZ body part. add a note that it is something to examine for future analysis - other sensors.

3.2 Visual Evidence for Fixed Effects

To determine which variables influence heat flow, we examined the measurements across different experimental conditions. The exploratory plots indicate that heat flow is not constant and is influenced by every categorical variable in the study.

Figure 1 displays heat flow during the sitting activity, partitioned by Day, Session, Sensor, and Sex.

For Sensor Type, we can see there are distinct vertical offsets between panels (e.g., HF_16 vs. HF_1), showing that different body locations have different baseline heat flow. Comparing the top row (Day 1, 24°C) to the bottom row (Day 2, 18°C) reveals a clear shift in mean heat flow, confirming a strong environmental effect. The solid linear fit lines across the four sessions often show non-zero slopes, suggesting that heat flow changes over the course of the day even when activity is held constant. The red (Female) and blue (Male) lines are frequently separated, indicating that biological sex influences thermal regulation.

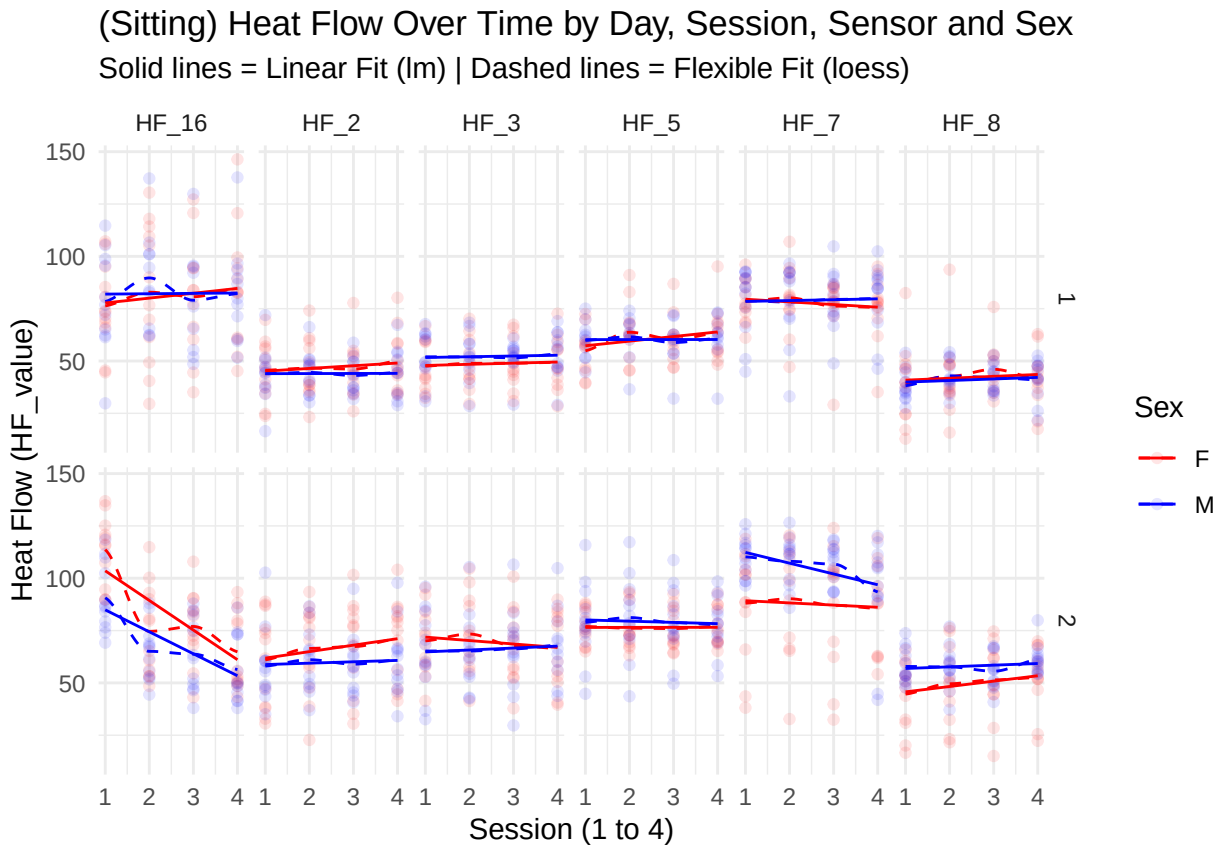


Figure 1: Sitting's Heat Flow Over Time by Day, Session, Sensor and Sex

Figure 2 focuses on Day 1 to highlight the impact of activity: Comparing the rows for sitting, standing, and walking reveals that heat flow varies with physical movement.

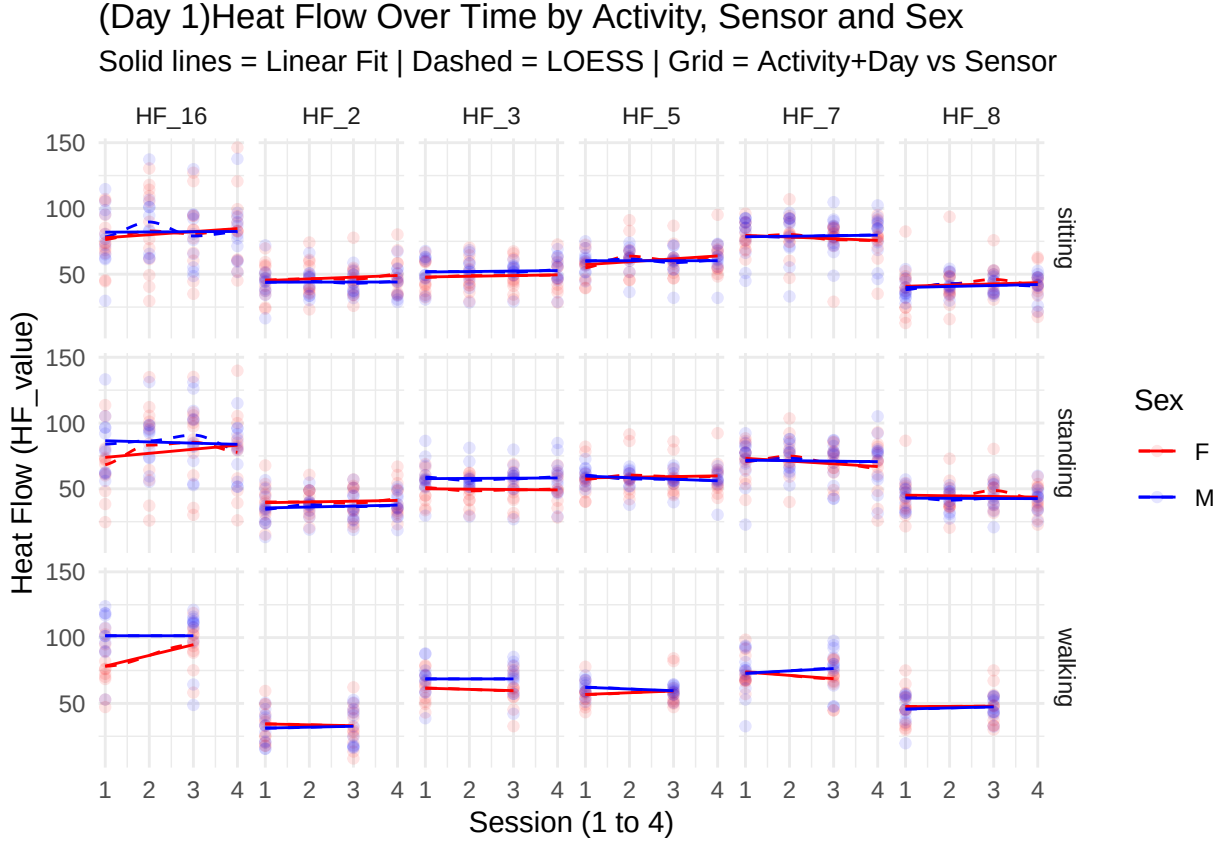


Figure 2: Day 1's Heat Flow Over Time by Activity, Sensor and Sex

Because the plots and physiological intuition suggest that every categorical variable exerts a fixed effect on the response, all predictors were carried forward into the modeling stage.

4 Statistical Description of the Methods

To analyze the factors influencing human heat flow while accounting for the complex hierarchical and temporal dependencies inherent to the experimental design, a Linear Mixed Model (LMM) was employed.

4.1 Fixed-Effects Optimization and Transformation

4.1.1 Nikita note: add note of what section the log stuff was shown, removed some description, discuss tests that zhuo ran

An initial naive ordinary least squares (OLS) regression was fitted to identify the baseline relationships between the predictors and the response variable and justify the inclusion of all 5 variables. Diagnostic plots of this initial model revealed significant heteroscedasticity and heavy-tailed residuals. To satisfy the assumptions of linear modeling and stabilize the variance, a natural logarithm transformation was applied to the heat flow measurements ($Y = \log(\text{HF_value})$).

4.1.2 Nikita note: discuss part F-tests. I think i used the likelihood ratio test, basically, from my anovas. I think that would be preferred - can we change this?

To determine the optimal fixed-effects structure, a systematic, multi-step model-building approach was employed. First, partial F-tests were utilized to evaluate the significance of predictors in a main-effects-only model.

For a given categorical predictor or interaction term with r degrees of freedom (corresponding to $k - 1$ coefficients for a variable with k levels), the partial F-test evaluates whether the inclusion of the term provides a statistically significant improvement to the model's goodness-of-fit. The null hypothesis posits that all r coefficients associated with the specific term are simultaneously zero:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_r = 0$$

To test this, the F-statistic is calculated by comparing the Residual Sum of Squares (RSS) of a reduced model (excluding the term in question) against a full model (including the term):

$$F = \frac{(RSS_{\text{reduced}} - RSS_{\text{full}})/r}{RSS_{\text{full}}/(n - p)}$$

(Where n represents the total number of observations, and p represents the total number of parameters estimated in the full model. Note that r is equivalent to $p_{\text{full}} - p_{\text{reduced}}$).

Under the null hypothesis, this test statistic follows an F-distribution with r and $n - p$ degrees of freedom ($F_{r, n-p}$). The mathematical validity of this test assumes that the model's residual errors are independent, identically distributed, and follow a normal distribution, $\epsilon \sim \mathcal{N}(0, \sigma^2)$.

4.1.3 nikita note: should we go through all of this? I think it is better to just explain what an interaction term is, and mention we used a similar methodology to the ANOVA done on the fixed effects... like, how is it diff?

Next, to identify relevant interaction terms, a preliminary marginal screening of ten theoretically plausible two-way interactions was conducted. To test whether the addition of each interaction term significantly improved the model (Null Hypothesis (H_0): The coefficient for the interaction term is exactly zero, $\beta_{\text{interaction}} = 0$), individual partial F-tests were performed.

4.1.4 Nikita note: I don't even look at the FWER.... we can mention later?

Because conducting multiple simultaneous hypothesis tests artificially inflates the probability of Type I errors (false positives), a Bonferroni correction was applied to control the family-wise error rate (FWER). Based on a standard family-wise alpha level of 0.05 distributed across 10 tests, the adjusted significance threshold was established at $p < 0.005$.

4.1.5 nikita note: again, i do not use partial F-test

To finalize the fixed-effects structure, a marginal backward elimination procedure (using partial F-test) was conducted for the resulting model. Furthermore, to adhere to the principle of marginality, any main effect involved in a significant two-way interaction was retained in the model, regardless of its independent main-effect significance.

4.2 Hierarchical Random-Effects Structure

The experimental design involved repeated measurements from the same 24 participants across multiple days, sessions, and physical activities. An assumption of standard linear regression is that all observations are independent. However, because multiple heat flow measurements were taken from the same individuals, unobserved subject-specific physiological traits (such as baseline metabolic rate or core temperature) were expected to cause intra-subject correlation.

To appropriately account for this unobserved heterogeneity and resolve the violation of independence, a Linear Mixed Model (LMM) was employed.

Mathematically, an LMM extends the general linear model by partitioning the total variance into fixed and random components. In matrix notation, the general model is defined as:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{b} + \epsilon$$

Where:

$\mathbf{y}$ is the vector of the response variable ($\log(\text{HF_value})$).

$\mathbf{X}$ is the design matrix for the fixed effects (the controlled experimental conditions such as Day, Session, and Activity).

β is the vector of fixed-effect coefficients, representing the population-average responses.

$\mathbf{Z}$ is the design matrix for the random effects. This matrix can contain both indicator variables for grouping factors (yielding random intercepts) and continuous/categorical covariates (yielding random slopes).

$\mathbf{b}$ is the vector of random effects. In this study, it captures both a subject's unique baseline shift (a random intercept) and their specific physiological trajectory in response to a changing experimental condition (a random slope). It is assumed to be normally distributed with mean zero and covariance matrix $\mathbf{G}$, such that $\mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathbf{G})$.

ϵ is the vector of residual errors, assumed to be normally distributed with mean zero and covariance matrix $\mathbf{R}$, such that $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$.

By introducing the random effects vector $\mathbf{b}$, the LMM defines the marginal covariance matrix of the response $\mathbf{y}$ as:

$$\text{Var}(\mathbf{y}) = \mathbf{Z}\mathbf{G}\mathbf{Z}^T + \mathbf{R}$$

Unlike standard Ordinary Least Squares (OLS) regression, which assumes a strictly diagonal covariance matrix where all observations are completely independent, this $\mathbf{Z}\mathbf{G}\mathbf{Z}^T$ structure mathematically permits non-zero covariances between observations belonging to the same participant. Furthermore, by allowing $\mathbf{G}$ to estimate both random intercepts and random slopes, the model accounts for the fact that participants not only start at different thermal baselines, but also adapt to the environmental temperature changes at entirely different physiological rates.

4.3 Residual Variance and Temporal Correlation

In standard linear mixed-effects models, the residual errors (ϵ_{ijk}) are conventionally assumed to be conditionally independent and identically distributed with a constant variance σ^2 . However, given the longitudinal nature of this experimental design, the assumption of residual independence may be overly restrictive. Physiologically, heat flow is a continuous, dynamic process; therefore, it is hypothesized that short-term, unexplained biological variations exhibit temporal autocorrelation. Specifically, a participant's unobserved physiological state during one session is likely to influence their state in the immediately subsequent session, with this residual correlation decaying as the time gap increases.

To mathematically account for this temporal decay across consecutive measurements, a first-order Autoregressive AR(1) correlation structure was implemented. This temporal memory was explicitly nested within the experimental day (person/HF_type/day). This boundary specification ensures that residual autocorrelation decays continuously across consecutive sessions within a single day, but is forced to assume complete independence across the overnight gap between Day 1 and Day 2.

Let ρ be the autoregressive parameter, and $|k - k'|$ be the chronological time lag between two measurement instances. Under AR(1) assumption, the covariance between any two residuals from the same sensor on the same person is defined as:

$$\text{Cov}(\epsilon_{ijk}, \epsilon_{ijk'}) = \sigma^2 \rho^{|k-k'|} I(\text{Day}_{ijk} = \text{Day}_{ijk'})$$

Where the indicator function $I(\cdot)$ returns 1 if the measurements occurred on the same day, and 0 if they occurred on different days, breaking the temporal correlation overnight.

While the AR(1) specification successfully captures temporal memory, it still imposes a strict assumption of homoscedasticity, forcing a single, universal residual variance (σ^2) across all observations. Physiologically, different sensor placements (e.g., wrist versus chest) experience varying degrees of ambient noise interference and physical contact stability.

To resolve this limitation, the model's residual structure was upgraded to an Autoregressive Heteroscedastic ARH(1) formulation. This structure retains the autoregressive temporal decay (ρ) but relaxes the constant variance assumption by calculating a unique, independent residual variance (σ_j^2) for each specific sensor type j .

The final residual errors are distributed as a vector $\epsilon_{ij} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{ij})$, where the elements of the block-diagonal residual covariance matrix $\mathbf{R}_{ij}$ are formalized as:

$$Cov(\epsilon_{ijk}, \epsilon_{ijk'}) = \sigma_j^2 \rho^{|k-k'|} I(\text{Day}_{ijk} = \text{Day}_{ijk'})$$

This final $\mathbf{R}$ matrix allows the baseline magnitude of the physiological noise to scale up or down depending on the specific physical sensor, while preserving the chronological decay of that heat flow noise as the experimental day progresses.

5 Nikita's Notes (outside of results)

- How to justify use of LMM - discuss more...
- Add coefficients for each equation, and discuss whether used ML or REML (and impacts?)
- Explain ICC
- When discussing ICCs, just create tables to show it?? Each row is a model, and each column is a variable?
- Add final summary table to show coefficients, etc of final model; add discussion!
- discuss adding activity to ar1 - better correlation?
- Explain AIC, BIC? likelihood ratio test stat? Log likelihood?

6 Results

6.1 Initial Model

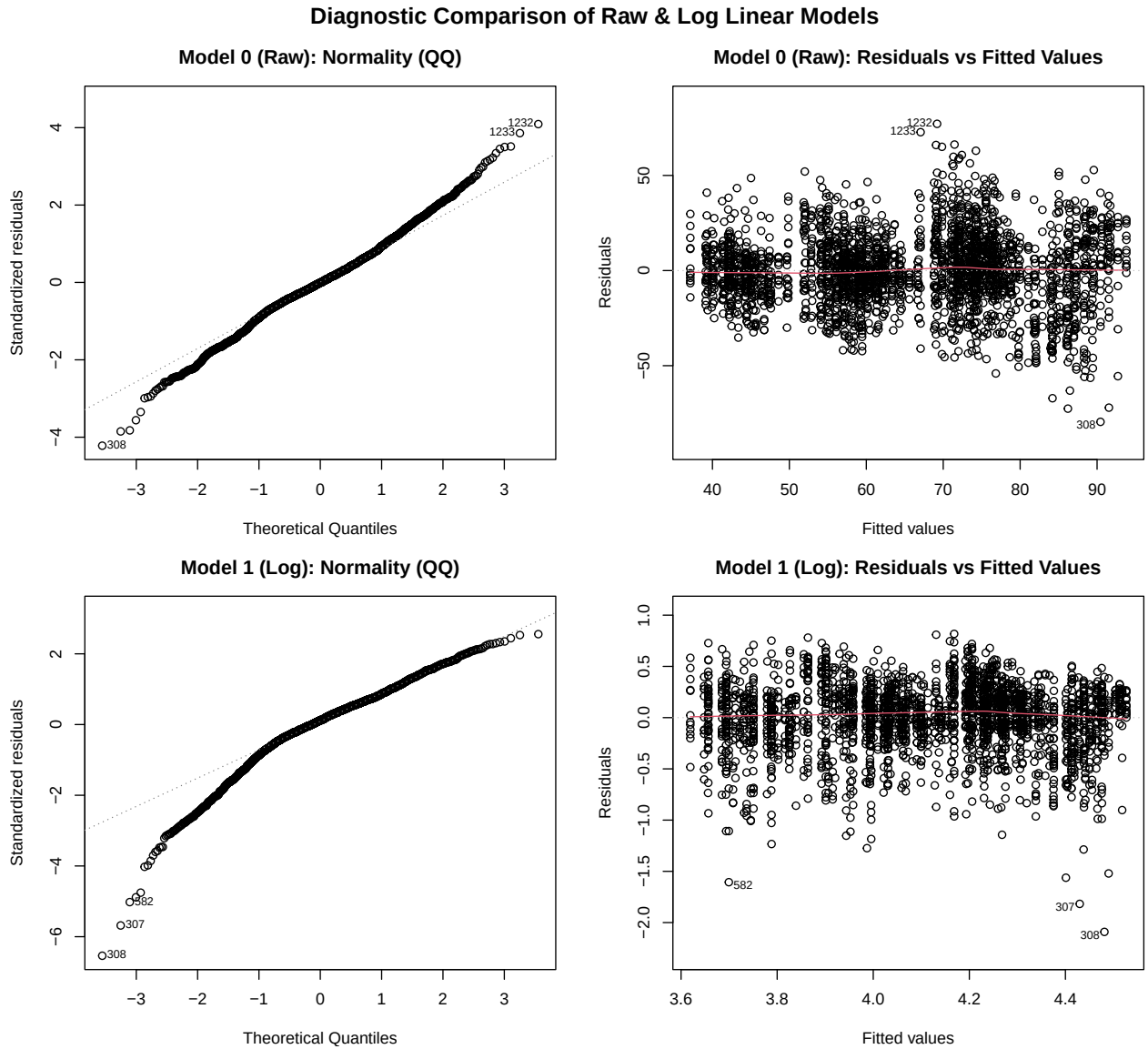


Figure 3: The QQ-plot of the Model 0 (the raw (untransformed) model) shows an approximately normal model with slightly heavy tails, but the Residuals vs Fitted Values plot shows heteroscedastic residuals, with variance increasing as fitted values increase. Model 1 (the log-transformed model) corrects the right heavy tail but the left tail deviates more, and the Scale-Location plot shows fairly homoscedastic residuals.

As seen in Figure 3, applying a simple linear model (Model 0: $\text{HF_value} \sim \text{session} + \text{activity} + \text{day} + \text{HF_type} + \text{sex}$) shows that the residuals are approximately normal (with heavy tails), but heteroscedastic. Transforming the response variable (Model 1: $\log(\text{HF_value})$) resulted in homoscedasticity and corrected the heavy right tail, but introduced more left-skew in the residuals. We prioritize homoscedasticity and use Model 1 (the log model) going forward, to ensure reliable parameter estimation and significance testing.

Table 1: Model selection of fixed effects via backwards elimination. Model 1 (the 'Full Model') includes all predictors. Each subsequent row represents a model where only one term was dropped from Model 1. The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the reduced model to Model 1. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Backwards Selection of Fixed Effects: Likelihood Ratio Test Summary						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 1 (Full Model)</i>	14	1499	1581	-736	–	–
Drop Day	13	1755	1832	-865	258.01	< 1e-04
Drop Session	11	1498	1563	-738	5.04	0.169116
Drop Activity	12	1506	1576	-741	10.78	0.004567
Drop Sensor	9	2438	2491	-1210	948.97	< 1e-04
Drop Sex	13	1506	1582	-740	8.77	0.003054

6.2 Fixed Effects

As seen in Table 1, although Model 1 includes all fixed effects ($\log(\text{HF_value}) \sim \text{session} + \text{activity} + \text{day} + \text{sensor} + \text{sex}$), a backwards selection process was conducted to ensure parsimony. We evaluated each variable by comparing the full model to nested models where one term was removed, using the Likelihood Ratio Test (LRT) and changes in information criteria (AIC and BIC).

At first glance, all variables are statistically significant except for the model dropping the session variable ($\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$), as the p-value from the likelihood ratio test of this model versus the full model is not below 0.05 ($p = 0.1691$). Additionally, although the models that drop activity and drop sex are technically statistically significant (p -values < 0.01), these models show a unique characteristic of having slightly higher AICs than the full model, but lower or similar BICs. This discrepancy implies that although these variables improve prediction, the improvement may not be large enough to justify including the variables. In this case (and similar cases further in the analysis), the variables are included but will be re-examined in final model pruning.

As such, the final fixed effect model was Model 2 ($\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$).

6.3 Interaction Terms

We further examined fixed effects by examining the value of adding all possible interaction terms to the model, as shown in Table 2. Testing adding each interaction term against against Model 2 ($\text{HF_value} \sim \text{activity} + \text{day} + \text{sensor} + \text{sex}$), we saw that adding the following interaction terms results in models with worse (or similar) AIC & BIC than the current model without interaction terms, and thus did not add them to the model: Activity & Day, Activity & Sex, Session & Sex, Day & Sex, Day & Session.

Technically, adding the interaction of Day & Session results in a p-value of 0.0118, below 0.05. However, this interaction increased the BIC and barely decreases the AIC, so we chose not to include it in the model. Note that the interaction of Activity & Session was not tested - because there is no data for certain interactions (ex. walking & session 2, walking & session 4), the model cannot run.

The interaction of Sensor with the other possible variables (Day, Activity, Sex, and Session) all have low p-values (<0.001) and result in decreased AIC, although two interaction terms (Sensor and Sex, and Sensor and Session) result in increased BIC. Table 1 showed a similar story - both Session and Sex decreased AIC slightly, but increased BIC. Similarly, we include these interaction terms for now and will prune the model at the end.

This results in Model 3: ($\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex} + \text{sensor} * \text{activity} + \text{sensor} * \text{day} + \text{sensor} * \text{sex} + \text{sensor} * \text{session} + \text{session}$). Note that session is added back into the model automatically, as the interaction of sensor and session is included, due to the principle of marginality. Table 3 shows the results of

Table 2: The top row (Model 2) is the row without any interaction terms. Each subsequent row represents a model where only one interaction term was added to Model 2. The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the new model to Model 2 via ANOVA. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Interaction Terms: Likelihood Ratio Test Summary						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 2 (No Interaction Terms)</i>	11	1498	1563	-738	–	–
Interact Sensor and Day	16	1311	1405	-640	196.89	< 1e-04
Interact Sensor and Activity	21	1340	1463	-649	178.67	< 1e-04
Interact Sensor and Sex	16	1482	1576	-725	25.78	< 1e-04
Interact Sensor and Session	29	1455	1625	-698	79.58	< 1e-04
Interact Activity and Day	13	1502	1578	-738	0.38	0.82859
Interact Activity and Sex	13	1499	1575	-736	3.43	0.17994
Interact Session and Sex	17	1504	1604	-735	6.53	0.36654
Interact Day and Sex	12	1500	1570	-738	0.63	0.42662
Interact Day and Session	17	1494	1594	-730	16.4	0.01177

checking if removing any of the four interaction terms improves Model 3. This showed a similar story to the individual tests of adding interaction terms—the inclusion of the interactions of Sensor and Sex, and Sensor and Session, slightly decrease AIC but increase BIC.

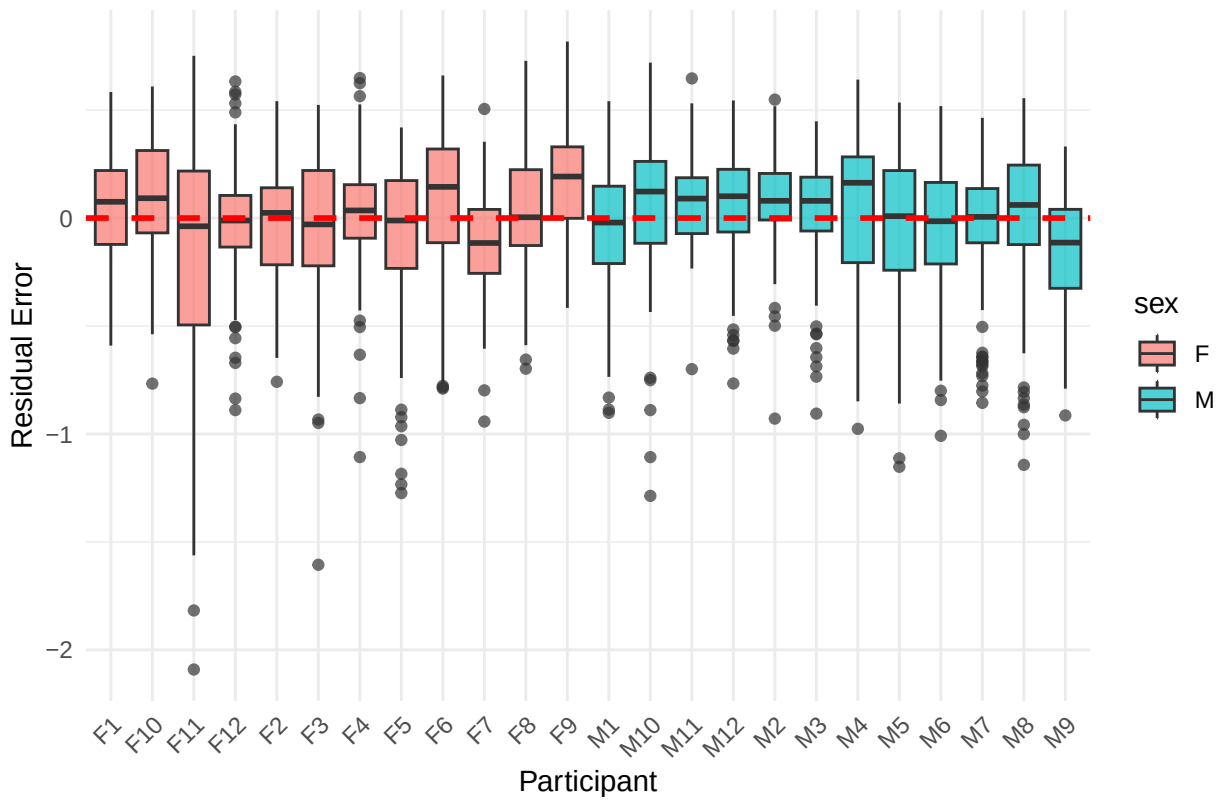
Table 3: Model selection of interaction terms via backwards elimination. Model 3 includes the four interaction terms previously noted. Each subsequent row represents a model where only one interaction term was dropped from Model 3. Note that for the last row, Drop Sensor and Session, the fixed effect Session is dropped as well (Session was only included in Model 3 because the interaction of Sensor and Session was added). The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the reduced model to Model 3. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Backwards Selection of Interaction Terms: Likelihood Ratio Test Summary						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 3 (Four Interaction Terms)</i>	49	1107	1395	-505	–	–
Drop Sensor and Day	44	1314	1573	-613	216.74	< 1e-04
Drop Sensor and Activity	39	1248	1477	-585	160.62	< 1e-04
Drop Sensor and Sex	44	1127	1386	-520	30.02	< 1e-04
Drop Sensor and Session	31	1113	1295	-525	41.49	0.001304

6.4 Random Intercepts

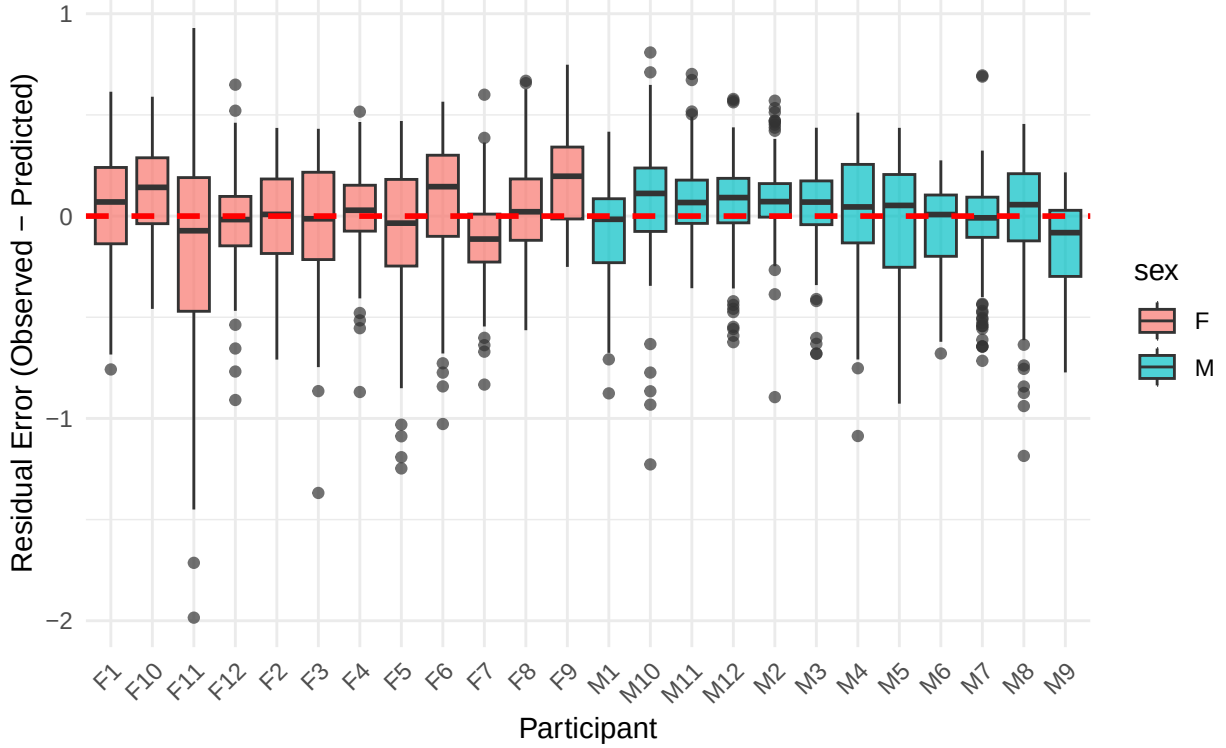
6.4.1 TO DO: ADD JUSTIFICATION FOR LMM

Residuals by Person for Model 1, fixed-effect-only model



Motivation for LMM: Residuals by Person (Model 3)

Systematic shifts away from zero prove that errors within an individual are correlated.



To determine if there is inter-individual variability, we added a random intercept for each person, creating Model 4: $\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sex} + \text{sensor} * \text{activity} + \text{sensor} * \text{day} + \text{sensor} * \text{sex} + \text{sensor} * \text{session} + \text{session}$. This assumes that all individuals have unique baselines for heat flow, regardless of if they are performing the same activity and/or are at the same temperature. Note the models in this section were run using REML to ensure estimates of variance components are not biased, unless explicitly stated otherwise.

Table 4: Impact of adding a random intercept for person (Model 4) to the fixed-effect only model (Model 3). Note this uses REML to compare models, as fixed effects are not changing. As such, the final p-values should be divided by 2. Given the p-value is already extremely low, this has no impact on the takeaway that we should include the Random Intercept for Person. Note models were fit using Restricted Maximum Likelihood (REML), as fixed effects are the same between models.

Likelihood Ratio Test for Random Intercept for Person						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 3	49	1356	1643	-629	—	—
Model 4	50	1208	1501	-554	149.96	< 1e-04

The Intraclass Correlation Coefficient (ICC) (also known to as intra-individual coefficient here) for Model 4 was 0.0791, implying that ~8% of the variance is due to differences between people (and 92% is unexplained). This suggests there is some inter-individual variability. Table 4 shows that this also reduces AIC and BIC significantly,

We then examined if a nested activity-level random effect was necessary. The ICC for person remains the same as in model 4 (0.0791), and the ICC of activity (within person) is 0, implying adding activity as a random intercept adds no value - the effect of activity does not vary randomly between people.

We then examined if a nested sensor-level random intercept was necessary (Model 5). The ICC for person is now 0.0184, lower than in model 4 (0.0791), and the ICC of sensor (within person) is 0.3772, with 60% of the variance unexplained. This implies that the major source of clustering is at the sensor level, and not individual level.

Removing the person random intercept (resulting in a sensor-only random intercept model) results in the significantly lower ICC for sensor (0.0159), with 98% of the variance left unexplained. This implies that the sensor by itself is not the primary driver of variance; instead, there is a shared baseline amongst sensors belonging to the same individual. Thus, we kept the nested sensor-level random intercept model (model 5).

6.5 Random Slopes

The current model has random intercepts for person and sensor (nested), implying that measurements from the same sensor are correlated, and sensors belonging to the same person are correlated. However, it assumes that the impact of temperature (the day variable) is identical for all individuals. We tested day as a random slope (given person) to determine if the effect of day varies by person, meaning some people adapt to the temperature environment differently than others.

Table 5: Likelihood test between Model 5 and Model 6, which adds day as a random slope. Model 6 has significantly lower AIC and BIC, and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Likelihood Ratio Test for Random Slope for day						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 5: Random Intercept: Person and Nested Sensor	51	485	784	-191	—	—
Model 6: Add Random Slope for Day (Given Person)	53	312	622	-103	176.82	< 1e-04

Adding day as a random slope (Model 6) results in 45% unexplained variance, compared to 60% from Model 5. The ICC for day is 0.186, which is fairly large; ~19% of the variation in heat flow measurements is explained by the fact that people have different heat flow reactions depending on the external temperature. Additionally, the correlation between the person-level intercept and the day-level random slope is -0.884. This implies that the higher a person’s heat flow was on day 1 (hot day), the steeper their drop was on day 2 (cold day), suggesting convergence.

The ICC for person also increased to 0.0419 (compared to 0.0184 from model 5), suggesting that the noise from people reacting differently to temperature masked their baseline differences. The ICC for sensor decreased to 0.3241 (from 0.3772 in model 5), suggesting that temperature explained some of the heat flow variance that was previously attributed to sensor.

Additionally as shown in Table 5, Model 6 has better AIC and BIC than Model 5, so we included day as a random slope.

6.6 Covariance Structures

Table 6: Likelihood test between Model 6 and Model 7, testing an AR1 structure where magnitude of correlation depends on amount of time between observations (instead of flat correlation). Model 7 has significantly lower AIC and BIC and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Likelihood Ratio Test for Covariance Structures						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 6: Assumes Compound Symmetry	53	312	622	-103	—	—
Model 7: Assumes Auto-regressive AR(1)	54	-402	-86	255	715.88	< 1e-04

The selected hierarchical structure (Model 6) effectively partitions the data’s variance into three distinct components: global participant-level baselines, individual thermal adaptation patterns (represented by the random slope for day), and nested sensor-level deviations. This framework accounts for the inherent dependencies in the data while acknowledging that heat flow sensitivity is not uniform across the study population.

However, this still imposes a strict assumption of conditional independence for the remaining residual errors. It has a compound symmetry structure, assuming a constant correlation structure where the correlation between session 1 and session 2 is identical to the correlation between session 1 and session 4, ignoring the fact that observations closer in time are typically more highly correlated. As such, we examined implementing an autoregressive (AR1) correlation structure for the residuals in Model 7. We assumed this was nested within person, sensor, and day, ensuring independence between people, sensor location, and days. We did not assume independence between activity because the activities were done in order, with session 1 consisting of sitting, then standing, then walking, then a break, and then session two beginning with the same cycle (although sessions 2 and 4 did not have walking). This assumes that these measurements were equally spaced in time, which is likely not true - timestamp data could improve this assumption.

The correlation structure has a ϕ of 0.5745, meaning that the correlation between session 1 and 2 is ~ 0.57 , and session 1 & 3 is $\sim 0.57^2$. This is fairly high, suggesting the previous data point strongly influences the next one. Since we have the data segmented by session and activity, it suggests that sitting in session 1 impacts standing in session 1, and walking in session 1 impacts sitting in session 2. Additionally, Table 6 shows that Model 7 has significantly lower AIC and BIC, and is statistically significant in the likelihood ratio test against Model 6, suggesting an AR1 covariance structure is a significant improvement on the Compound Symmetry structure, and that there is a strong temporal auto-correlation.

Table 7: Likelihood test between Model 7 and Model 8, testing an ARH1 structure that allows each sensor to have its own unique residual variance instead of assuming uniform residual noise. Model 8 has significantly lower AIC and BIC and a very small p-value, so it is statistically significant. Note that models are run using REML, as they have the same fixed effects.

Likelihood Ratio Test for Covariance Structures						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
Model 7: Assumes Auto-regressive Structure (AR(1))	54	-402	-86	255	–	–
Model 8: Assumes Heterogeneous AR Structure (ARH(1))	59	-944	-598	531	551.73	< 1e-04

Model 7 (AR(1)) showed that there is a temporal aspect to the data, where consecutive sessions are more closely linked than further ones. However, AR(1) assumes that the amount of variance is identical for all sensors. However, we would expect sensors on different body parts to have different variance. As such, we tested a heterogeneous AR structure ARH(1) by combining Model 7’s AR(1) structure with heterogeneous residual variances across sensors.

Model 8’s ϕ is 0.6269, higher than Model 7’s (0.5745), suggesting the temporal aspect of the data was higher than previously suggested and was masked by the noise levels of the sensors. The estimates for the heterogeneous variances relative to sensor 16 (baseline, dominant hand) are: 1.27 (sensor 2, chest (side)), 0.72 (sensor 3, upper chest), 0.35 (sensor 5, upper arm), 0.95 (sensor 7, lower arm), and 0.97 (sensor 8, abdomen). This implies that sensors on the abdomen, lower arm, and dominant hand have similar noise. The sensor on the chest (side) is significantly noisier than those sensors, and the sensors on the upper chest and arm are significantly less noisy. There does not seem to be an obvious trend or reason for this; it may be helpful to consult experts in the field. Additionally, Table 7 shows that the ARH(1) structure (Model 8) has significantly lower AIC and BIC than Model 7’s AR(1) structure, and is statistically significant in the likelihood ratio test.

6.7 Pruning Fixed Effects

The final model utilizes an ARH(1) covariance structure to account for the temporal aspect of the data between sequential activities and sessions, while allowing for unique levels of noise at each sensor. Additionally, it has random intercepts for individuals and nested sensors to account for sensor-specific baselines

per individual, and a person-level random slope for day to capture individual level responses to temperature changes. The model has become quite complex, so we re-examine fixed effects to determine if there are any that can be removed.

Table 8: Model selection of variables via backwards elimination. Model 8 includes the four interaction terms previously noted. Each subsequent row represents a model where either only one interaction term was dropped from Model 8 (Drop X and Y Interaction), or the whole variable and interaction (Drop Sex Fully means dropped the variable sex, and the interaction term sex and sensor). The LRT Statistic and P-Value are derived from Likelihood Ratio Tests comparing the reduced models to Model 8. Note all models were fit using Maximum Likelihood (ML), to compare across models.

Backwards Selection of Interaction Terms and Variables: Likelihood Ratio Test Summary						
Model Version	Degrees of Freedom	AIC	BIC	Log-Likelihood	Likelihood Ratio Test Stat	P-Value
<i>Model 8 (Full Model)</i>	59	-1193	-846	655	–	–
Drop Sensor and Sex Interaction	54	-1198	-881	653	4.13	0.530570
Drop Sex Fully	53	-1199	-887	652	5.56	0.473754
Drop Sensor and Day Interaction	54	-1135	-818	622	67.53	< 1e-04
Drop Day Fully	53	-1107	-795	606	97.93	< 1e-04
Drop Sensor and Activity Interaction	49	-809	-521	453	403.85	< 1e-04
Drop Activity Fully	47	-769	-493	432	447.07	< 1e-04
Drop Sensor and Session Interaction	44	-1187	-928	638	35.49	0.002093
Drop Session Fully	41	-1190	-949	636	38.39	0.003439

Table 8 suggests that we should remove the variables Sex and the interaction term of Sensor and Sex, as the model without those terms has better AIC and BIC, and the p-value from the likelihood ratio test is not statistically significant (>0.4). We also removed the variable Session and the interaction term of Sensor and Session. Although they minimally improve the AIC (by <10 points), they worsen the BIC significantly and introduce ~ 18 degrees of freedom.

The final model is Model 9: $(\log(\text{HF_value}) \sim \text{activity} + \text{day} + \text{sensor} + \text{sensor} * \text{activity} + \text{sensor} * \text{day})$, which evaluates the fixed effects of activity, day, sensor, and the interactions of sensor with activity and day. It addresses the hierarchical nature of the data by incorporating a random intercept for person, a nested random intercept for sensor, and a person-level random slope for day. Finally, the error structure is defined by an ARH(1) covariance structure, allowing temporal auto-correlation segmented by person, sensor, and day, and heterogenous residual variances to account for different levels of volatility across sensors.

Linear mixed-effects model fit by REML

```
Data: data_clean
      AIC      BIC   logLik
-1037.279 -796.7589 559.6394
```

Random effects:

```
Formula: ~1 + day | person
Structure: General positive-definite, Log-Cholesky parametrization
      StdDev      Corr
(Intercept) 0.08198097 (Intr)
day2         0.13882909 -0.922
```

```
Formula: ~1 | HF_type %in% person
      (Intercept) Residual
StdDev:   0.1591161 0.2849547
```

Correlation Structure: AR(1)

```
Formula: ~1 | person/HF_type/day
Parameter estimate(s):
      Phi
```

0.6324753

Variance function:

Structure: Different standard deviations per stratum

Formula: ~1 | HF_type

Parameter estimates:

	HF_16	HF_2	HF_3	HF_5	HF_7	HF_8
	1.0000000	1.2388910	0.7058033	0.3412306	0.9310399	0.9463243

Fixed effects: log(HF_value) ~ (day + activity + sex) * HF_type

	Value	Std.Error	DF	t-value	p-value
(Intercept)	4.382541	0.06525206	2476	67.16326	0.0000
day2	-0.146868	0.05693764	2476	-2.57945	0.0100
activitystanding	-0.077875	0.01749847	2476	-4.45040	0.0000
activitywalking	0.094058	0.02127396	2476	4.42128	0.0000
sexM	-0.003530	0.08263706	22	-0.04272	0.9663
HF_typeHF_2	-0.570742	0.09382998	110	-6.08273	0.0000
HF_typeHF_3	-0.538773	0.08400422	110	-6.41365	0.0000
HF_typeHF_5	-0.317337	0.07947313	110	-3.99301	0.0001
HF_typeHF_7	-0.116802	0.08715457	110	-1.34018	0.1830
HF_typeHF_8	-0.781547	0.08765932	110	-8.91573	0.0000
day2:HF_typeHF_2	0.470108	0.07826288	2476	6.00679	0.0000
day2:HF_typeHF_3	0.467161	0.06086916	2476	7.67484	0.0000
day2:HF_typeHF_5	0.412815	0.05278869	2476	7.82013	0.0000
day2:HF_typeHF_7	0.308993	0.06682584	2476	4.62386	0.0000
day2:HF_typeHF_8	0.464246	0.06796492	2476	6.83067	0.0000
activitystanding:HF_typeHF_2	-0.105925	0.02754265	2476	-3.84585	0.0001
activitywalking:HF_typeHF_2	-0.461892	0.03378722	2476	-13.67060	0.0000
activitystanding:HF_typeHF_3	0.178397	0.02139428	2476	8.33853	0.0000
activitywalking:HF_typeHF_3	0.170678	0.02612520	2476	6.53307	0.0000
activitystanding:HF_typeHF_5	0.051458	0.01850259	2476	2.78110	0.0055
activitywalking:HF_typeHF_5	-0.110354	0.02252214	2476	-4.89980	0.0000
activitystanding:HF_typeHF_7	-0.037174	0.02362961	2476	-1.57320	0.1158
activitywalking:HF_typeHF_7	-0.144755	0.02890232	2476	-5.00842	0.0000
activitystanding:HF_typeHF_8	0.162279	0.02399148	2476	6.76404	0.0000
activitywalking:HF_typeHF_8	0.091377	0.02923513	2476	3.12559	0.0018
sexM:HF_typeHF_2	-0.072653	0.12069664	110	-0.60195	0.5484
sexM:HF_typeHF_3	0.069682	0.11020538	110	0.63229	0.5285
sexM:HF_typeHF_5	0.019993	0.10593391	110	0.18873	0.8507
sexM:HF_typeHF_7	0.130835	0.11358630	110	1.15186	0.2519
sexM:HF_typeHF_8	0.093280	0.11428393	110	0.81621	0.4161

Correlation:

	(Intr)	day2	actvtys	actvtyw	sexM	HF_HF_2
day2	-0.433					
activitystanding	-0.134	0.013				
activitywalking	-0.097	-0.001	0.408			
sexM	-0.618	-0.019	-0.001	0.001		
HF_typeHF_2	-0.643	0.219	0.093	0.067	0.419	
HF_typeHF_3	-0.718	0.245	0.104	0.075	0.468	0.500
HF_typeHF_5	-0.759	0.259	0.110	0.079	0.494	0.528
HF_typeHF_7	-0.692	0.236	0.101	0.072	0.451	0.481
HF_typeHF_8	-0.688	0.235	0.100	0.072	0.448	0.479
day2:HF_typeHF_2	0.229	-0.547	-0.009	0.001	0.014	-0.402
day2:HF_typeHF_3	0.295	-0.704	-0.012	0.001	0.018	-0.206
day2:HF_typeHF_5	0.340	-0.812	-0.014	0.001	0.022	-0.238
day2:HF_typeHF_7	0.269	-0.641	-0.011	0.001	0.017	-0.187

day2:HF_typeHF_8	0.264	-0.630	-0.011	0.001	0.017	-0.185
activitystanding:HF_typeHF_2	0.085	-0.008	-0.635	-0.259	0.001	-0.147
activitywalking:HF_typeHF_2	0.061	0.000	-0.257	-0.630	-0.001	-0.105
activitystanding:HF_typeHF_3	0.110	-0.010	-0.818	-0.334	0.001	-0.076
activitywalking:HF_typeHF_3	0.079	0.001	-0.332	-0.814	-0.001	-0.055
activitystanding:HF_typeHF_5	0.127	-0.012	-0.946	-0.386	0.001	-0.088
activitywalking:HF_typeHF_5	0.091	0.001	-0.385	-0.945	-0.001	-0.063
activitystanding:HF_typeHF_7	0.099	-0.009	-0.741	-0.302	0.001	-0.069
activitywalking:HF_typeHF_7	0.071	0.001	-0.300	-0.736	-0.001	-0.049
activitystanding:HF_typeHF_8	0.098	-0.009	-0.729	-0.297	0.001	-0.068
activitywalking:HF_typeHF_8	0.070	0.001	-0.297	-0.728	-0.001	-0.049
sexM:HF_typeHF_2	0.412	0.013	0.001	-0.001	-0.668	-0.625
sexM:HF_typeHF_3	0.452	0.015	0.001	-0.001	-0.731	-0.314
sexM:HF_typeHF_5	0.470	0.015	0.001	-0.001	-0.761	-0.326
sexM:HF_typeHF_7	0.438	0.014	0.001	-0.001	-0.709	-0.305
sexM:HF_typeHF_8	0.435	0.014	0.001	-0.001	-0.705	-0.302
	HF_HF_3	HF_HF_5	HF_HF_7	HF_HF_8	d2:HF_HF_2	

day2						
activitystanding						
activitywalking						
sexM						
HF_typeHF_2						
HF_typeHF_3						
HF_typeHF_5	0.590					
HF_typeHF_7	0.538	0.568				
HF_typeHF_8	0.535	0.565	0.515			
day2:HF_typeHF_2	-0.179	-0.189	-0.172	-0.172		
day2:HF_typeHF_3	-0.358	-0.244	-0.221	-0.221	0.514	
day2:HF_typeHF_5	-0.265	-0.325	-0.255	-0.255	0.594	
day2:HF_typeHF_7	-0.209	-0.221	-0.368	-0.200	0.466	
day2:HF_typeHF_8	-0.206	-0.218	-0.198	-0.374	0.461	
activitystanding:HF_typeHF_2	-0.066	-0.070	-0.064	-0.063	0.014	
activitywalking:HF_typeHF_2	-0.047	-0.050	-0.046	-0.045	0.000	
activitystanding:HF_typeHF_3	-0.128	-0.090	-0.082	-0.082	0.007	
activitywalking:HF_typeHF_3	-0.091	-0.065	-0.059	-0.059	0.000	
activitystanding:HF_typeHF_5	-0.099	-0.116	-0.095	-0.094	0.009	
activitywalking:HF_typeHF_5	-0.071	-0.084	-0.068	-0.068	-0.001	
activitystanding:HF_typeHF_7	-0.077	-0.082	-0.136	-0.074	0.007	
activitywalking:HF_typeHF_7	-0.055	-0.058	-0.097	-0.053	0.000	
activitystanding:HF_typeHF_8	-0.076	-0.080	-0.073	-0.137	0.007	
activitywalking:HF_typeHF_8	-0.055	-0.058	-0.053	-0.098	0.000	
sexM:HF_typeHF_2	-0.320	-0.338	-0.309	-0.306	-0.022	
sexM:HF_typeHF_3	-0.646	-0.370	-0.338	-0.336	-0.012	
sexM:HF_typeHF_5	-0.364	-0.653	-0.352	-0.349	-0.013	
sexM:HF_typeHF_7	-0.340	-0.360	-0.644	-0.326	-0.010	
sexM:HF_typeHF_8	-0.338	-0.357	-0.326	-0.636	-0.011	
	d2:HF_HF_3	d2:HF_HF_5	d2:HF_HF_7	d2:HF_HF_8		

day2	
activitystanding	
activitywalking	
sexM	
HF_typeHF_2	
HF_typeHF_3	
HF_typeHF_5	

HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5	0.763			
day2:HF_typeHF_7	0.600	0.692		
day2:HF_typeHF_8	0.592	0.684	0.537	
activitystanding:HF_typeHF_2	0.007	0.009	0.007	0.007
activitywalking:HF_typeHF_2	0.000	-0.001	0.000	0.000
activitystanding:HF_typeHF_3	0.015	0.011	0.009	0.009
activitywalking:HF_typeHF_3	-0.001	-0.001	-0.001	-0.001
activitystanding:HF_typeHF_5	0.011	0.014	0.010	0.010
activitywalking:HF_typeHF_5	-0.001	-0.001	-0.001	-0.001
activitystanding:HF_typeHF_7	0.009	0.010	0.014	0.008
activitywalking:HF_typeHF_7	0.000	-0.001	0.000	0.000
activitystanding:HF_typeHF_8	0.009	0.010	0.008	0.014
activitywalking:HF_typeHF_8	0.000	-0.001	0.000	0.000
sexM:HF_typeHF_2	-0.014	-0.017	-0.011	-0.013
sexM:HF_typeHF_3	-0.014	-0.019	-0.013	-0.014
sexM:HF_typeHF_5	-0.016	-0.022	-0.013	-0.015
sexM:HF_typeHF_7	-0.013	-0.015	-0.013	-0.012
sexM:HF_typeHF_8	-0.015	-0.018	-0.012	-0.022
	actvtys:HF_HF_2	actvtyw:HF_HF_2	actvtys:HF_HF_3	
day2				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2	0.405			
activitystanding:HF_typeHF_3	0.520	0.210		
activitywalking:HF_typeHF_3	0.211	0.513	0.407	
activitystanding:HF_typeHF_5	0.601	0.243	0.774	
activitywalking:HF_typeHF_5	0.245	0.595	0.315	
activitystanding:HF_typeHF_7	0.470	0.190	0.606	
activitywalking:HF_typeHF_7	0.191	0.463	0.245	
activitystanding:HF_typeHF_8	0.463	0.187	0.597	
activitywalking:HF_typeHF_8	0.189	0.458	0.243	
sexM:HF_typeHF_2	0.000	0.001	0.000	
sexM:HF_typeHF_3	0.000	0.001	0.000	
sexM:HF_typeHF_5	0.000	0.001	-0.001	
sexM:HF_typeHF_7	0.000	0.001	-0.001	
sexM:HF_typeHF_8	0.000	0.001	0.000	
	actvtyw:HF_HF_3	actvtys:HF_HF_5	actvtyw:HF_HF_5	
day2				

activitystanding			
activitywalking			
sexM			
HF_typeHF_2			
HF_typeHF_3			
HF_typeHF_5			
HF_typeHF_7			
HF_typeHF_8			
day2:HF_typeHF_2			
day2:HF_typeHF_3			
day2:HF_typeHF_5			
day2:HF_typeHF_7			
day2:HF_typeHF_8			
activitystanding:HF_typeHF_2			
activitywalking:HF_typeHF_2			
activitystanding:HF_typeHF_3			
activitywalking:HF_typeHF_3			
activitystanding:HF_typeHF_5	0.314		
activitywalking:HF_typeHF_5	0.769	0.407	
activitystanding:HF_typeHF_7	0.246	0.700	0.285
activitywalking:HF_typeHF_7	0.599	0.284	0.695
activitystanding:HF_typeHF_8	0.242	0.690	0.281
activitywalking:HF_typeHF_8	0.593	0.281	0.687
sexM:HF_typeHF_2	0.001	-0.001	0.001
sexM:HF_typeHF_3	0.000	-0.001	0.001
sexM:HF_typeHF_5	0.001	-0.001	0.001
sexM:HF_typeHF_7	0.001	-0.001	0.001
sexM:HF_typeHF_8	0.001	-0.001	0.001
actvtys:HF_HF_7 actvtyw:HF_HF_7 actvtys:HF_HF_8			
day2			
activitystanding			
activitywalking			
sexM			
HF_typeHF_2			
HF_typeHF_3			
HF_typeHF_5			
HF_typeHF_7			
HF_typeHF_8			
day2:HF_typeHF_2			
day2:HF_typeHF_3			
day2:HF_typeHF_5			
day2:HF_typeHF_7			
day2:HF_typeHF_8			
activitystanding:HF_typeHF_2			
activitywalking:HF_typeHF_2			
activitystanding:HF_typeHF_3			
activitywalking:HF_typeHF_3			
activitystanding:HF_typeHF_5			
activitywalking:HF_typeHF_5			
activitystanding:HF_typeHF_7			
activitywalking:HF_typeHF_7	0.408		
activitystanding:HF_typeHF_8	0.540	0.219	
activitywalking:HF_typeHF_8	0.220	0.536	0.406
sexM:HF_typeHF_2	0.000	0.001	0.000

sexM:HF_typeHF_3	0.000	0.001	0.000	
sexM:HF_typeHF_5	0.000	0.001	0.000	
sexM:HF_typeHF_7	0.000	0.001	0.000	
sexM:HF_typeHF_8	0.000	0.001	0.000	
	actvtyw:HF_HF_8	sM:HF_HF_2	sM:HF_HF_3	sM:HF_HF_5
day2				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2				
activitystanding:HF_typeHF_3				
activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				
activitywalking:HF_typeHF_5				
activitystanding:HF_typeHF_7				
activitywalking:HF_typeHF_7				
activitystanding:HF_typeHF_8				
activitywalking:HF_typeHF_8				
sexM:HF_typeHF_2	0.001			
sexM:HF_typeHF_3	0.001	0.502		
sexM:HF_typeHF_5	0.001	0.522	0.572	
sexM:HF_typeHF_7	0.001	0.486	0.532	0.553
sexM:HF_typeHF_8	0.001	0.484	0.530	0.551
	sM:HF_HF_7			
day2				
activitystanding				
activitywalking				
sexM				
HF_typeHF_2				
HF_typeHF_3				
HF_typeHF_5				
HF_typeHF_7				
HF_typeHF_8				
day2:HF_typeHF_2				
day2:HF_typeHF_3				
day2:HF_typeHF_5				
day2:HF_typeHF_7				
day2:HF_typeHF_8				
activitystanding:HF_typeHF_2				
activitywalking:HF_typeHF_2				
activitystanding:HF_typeHF_3				
activitywalking:HF_typeHF_3				
activitystanding:HF_typeHF_5				

```

activitywalking:HF_typeHF_5
activitystanding:HF_typeHF_7
activitywalking:HF_typeHF_7
activitystanding:HF_typeHF_8
activitywalking:HF_typeHF_8
sexM:HF_typeHF_2
sexM:HF_typeHF_3
sexM:HF_typeHF_5
sexM:HF_typeHF_7
sexM:HF_typeHF_8          0.513

Standardized Within-Group Residuals:
      Min      Q1      Med      Q3      Max
-5.8180449 -0.4262270  0.1072506  0.5857570  3.2774555

Number of Observations: 2638
Number of Groups:
      person HF_type %in% person
      24      144

```

7 Discussion: Answer the questions

- Motivate the use of mixed linear models in this project and build one to assess
 - the inter-variability in heat flow,
 - * ??
 - the intra-variability in heat flow, and
 - * ??
 - the covariance structure between the sensors
 - * ??
- Allow for different levels of grouping: allow observations for the same participant to be correlated, and observations within the same participant and the same activity to be correlated too. Check whether this is needed by looking at the proportion of variance explained by your different levels
 - Person yes (prev discussed in ICC)
 - Activity no (prev discussed in ICC)
 - Sensor yes (prev discussed)
- The context of the experiment is multivariate by nature. How will you use the univariate mixed model to model marginal and joint measurements from the different sensors?
- Provide interpretation of the most important parameters of your final model
- Aim of the study: Is there significant inter- and intra-individual variability in heat flow measurements in each of the sensors? How are measurements from different sensors related?
- To examine the intra-individual variability, it is necessary to analyze for the same person
 - Effect of the time: The difference between the same activities on the same day, but at different times (sessions)
 - Temperature effect: The difference between the same activities but on different days
- To examine the inter-individual variability, it is necessary to examine the difference between the same activities on the same day, but across participants

7.1 Appendix